Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

Given the de Broglie wavelength of an electron is related to its momentum by Î» = h / p , where Î» is the wavelength , h is Planck's constant , and p is the momentum . We can calculate the momentum using the kinetic energy of the electron . First , we need to convert the kinetic energy from eV to Joules . Then we can find the momentum , and subsequently the wavelength .

</think>

<answer> F . 5 . 27 ÃĹ 10 ^-9 m